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Introduction to Deep Learning – Final Project Report 2026–2027

Explainable Retinal Image Classification

Glaucoma Severity Classification with XAI

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1. Abstract

This project investigates explainable glaucoma stage classification from retinal fundus images using deep learning. We used the ‘moondream/glaucoma-detection’ dataset to compare a custom CNN trained from scratch with ImageNet-pretrained DenseNet121 and ResNet50 models for classifying images as normal, early, or advanced. The pipeline included image preprocessing, data augmentation during training, and evaluation using accuracy, macro F1-score, and confusion matrices. In the current experiments, both pretrained models outperformed the custom CNN, suggesting an advantage for transfer learning under the tested configurations, while early-stage classification remained challenging. Following model evaluation, we applied three explainable artificial intelligence (XAI) methods: LayerCAM, GradCAM++, and Occlusion to a selected ResNet50 model. Their visual explanations were compared across correct and incorrect predictions to investigate influential image regions and possible causes of classification errors. This analysis complemented quantitative performance evaluation with qualitative insights into model behavior. However, unresolved image duplication across dataset splits limits evaluation reliability and requires correction before drawing firm conclusions about generalization.

2. Introduction and Research Question

Glaucoma severity classification from retinal fundus images involves distinguishing normal, early, and advanced categories using visual information from structures such as the optic disc and optic cup. Distinguishing these categories can be challenging because visual differences between stages may be subtle.

This project compared a custom CNN trained from scratch with ImageNet-pretrained DenseNet121 and ResNet50 models. Transfer learning enables the reuse of previously learned visual features, motivating an evaluation of its effectiveness compared with training from scratch on a limited dataset.

Beyond classification performance, we investigated which image regions influenced model predictions. High accuracy alone does not establish whether a model relies on relevant anatomical structures or unrelated features. Therefore, we applied LayerCAM, GradCAM++, and Occlusion to a selected ResNet50 model, comparing visual explanations for correct and incorrect predictions to examine model behavior and possible causes of classification errors. These explanations support qualitative analysis rather than establish clinical validity.

Research Question: Can transfer-learning-based CNNs outperform a CNN trained from scratch in three-class glaucoma severity classification, and how do LayerCAM, GradCAM++, and Occlusion differ in highlighting clinically relevant regions, particularly the optic disc and optic cup, for correct and incorrect predictions?

3. Related Work

Previous studies have applied CNN-based approaches for glaucoma detection and severity assessment. Latha and Aruna Priya [1] proposed a two-stage framework that first distinguishes normal from glaucomatous images, then classifies glaucoma as early or advanced using a Dual Functional CNN and features from the segmented neuroretinal rim. This provides a relevant comparison with direct three-class classification without an explicit optic disc/cup segmentation stage.

Beyond architecture selection, transfer learning is another important consideration in medical image classification. Kim et al. [2] reviewed pretrained CNN approaches, including feature extraction and fine-tuning, highlighting their usefulness when labelled data are limited. They also reported no consensus on an optimal fine-tuning configuration, supporting empirical evaluation of pretrained models for each target task.

In parallel with classification performance, explainability is also applied as a means of examining the basis of model predictions. Abd El-Gawad et al. [3] reviewed AI methods for glaucoma detection and classification, discussing explanation techniques such as Grad-CAM and emphasizing the need for clinically meaningful interpretations. This motivates examining visual explanations for both correct and incorrect predictions, while recognizing that heatmaps alone do not establish clinical validity.

Building on this literature, our project combines a direct comparison of three classification models with a qualitative comparison of three explanation methods, linking predictive performance with an investigation of model behavior.

4. Dataset and Data Preparation

4.1 Dataset

Dataset: Glaucoma Detection, Hugging Face Datasets Hub [4]. The dataset is split by the authors into train, validation, and test subsets:

Table 1: Dataset sizes before and after within-split deduplication.

Split	Original	After deduplication
Train	2,847	2,341
Validation	1,259	1,163
Test	1,272	1,221
Total	5,378	4,725

4.2 Data Preparation

Images are deduplicated within each split using MD5 hashes of their raw bytes, yielding the sizes in Table 1. Cross-split checks also identify 459 shared image hashes between training and validation, and 358 between training and test. These overlaps remain unre-

solved and limit the reliability of the evaluation.

Image preprocessing and augmentation. The following steps are applied before model training and evaluation:

- **Input:** RGB images resized to 224×224 for CNN/DenseNet121 or 320×320 for ResNet50.
- **Augmentation:** training images use horizontal flipping ($p = 0.5$), rotation ($\pm 10^\circ$), and color jitter (brightness/contrast: 0.1; saturation: 0.05). No random augmentation is applied during validation or testing.
- **Normalization:** ImageNet mean (0.485, 0.456, 0.406) and standard deviation (0.229, 0.224, 0.225).
- **Label encoding:** normal = 0, early = 1, advanced = 2.

5. Methods

This study compares a CNN trained from scratch with ImageNet-pretrained DenseNet121 and ResNet50 models. Three post-hoc XAI methods are applied to the selected ResNet50 model to examine the regions influencing its predictions.

5.1 Baseline CNN Method

The baseline is a randomly initialized CNN with five blocks containing 32, 64, 128, 256, and 512 channels. Each block comprises two bias-free 3×3 convolutions (stride 1, padding 1), each followed by batch normalization and ReLU, then 2×2 max-pooling with stride 2. Global average pooling, dropout ($p = 0.5$), and a linear layer produce three logits. The network contains 4,715,747 trainable parameters.

Training uses Adam and weighted cross-entropy with $w_c = N/(3n_c)$, where N is the training-set size and n_c is the class count after within-split deduplication. The checkpoint with the lowest validation loss is retained. Training settings are summarized in Section 6.1.

5.2 Main Method

ResNet50 is selected for explanation using LayerCAM, Grad-CAM++, and Occlusion. The rationale for this selection, including the trade-off between test performance and observed training behavior, is discussed in Section 7.2.

5.2.1 Selected Transfer-Learning Model

DenseNet121 [5] and ResNet50 [6] use ImageNet-pretrained backbones with their original classifiers replaced by three-output linear layers. DenseNet121 concatenates preceding feature maps within dense blocks, whereas ResNet50 uses additive residual shortcuts. ResNet50 is selected for XAI analysis; DenseNet121 remains a classification comparator. The selection rationale is discussed in Section 7.2.

5.2.2 Fine-Tuning Strategy

Both models first train only their classifiers for five epochs, then unfreeze the entire network. ResNet50 preserves backbone batch-normalization statistics during warm-up and uses a lower learning rate for the backbone during fine-tuning. DenseNet121 uses a shared fine-tuning learning rate. Checkpoints are selected using validation loss. Training budgets, optimization settings, and seeds are reported in Section 6.1.

5.2.3 Explainability Methods

Three post-hoc methods, LayerCAM, Grad-CAM++, and Occlusion, are used to investigate the image regions influencing ResNet50 predictions. LayerCAM and Grad-CAM++ use gradients and internal feature activations, whereas Occlusion measures the effect of perturbing the input. Their comparison provides complementary views of model behaviour rather than assuming that a single heatmap fully explains a prediction.

All methods use the selected ResNet50 checkpoint from seed 2026 in evaluation mode, without updating model weights. Explanations target the predicted class, including when it differs from the ground-truth label. This ensures that the maps examine the decision actually made by the model. Inputs follow the same 320×320 resizing and ImageNet normalization used during ResNet50 evaluation.

LayerCAM [7] computes gradients of the target-class score with respect to feature activations at a chosen layer. Positive gradients provide spatially varying weights for the corresponding activations, which are aggregated across channels to generate a class-specific map. This preserves location-dependent weighting rather than assigning one uniform importance weight to an entire feature map.

LayerCAM is included to inspect where localized evidence supporting a glaucoma-stage prediction appears in the image. The implementation uses the final residual block of ResNet50’s last residual stage as its target. Only this block is used; no fusion of explanations from multiple layers is performed.

Grad-CAM++ [8] derives channel importance weights from a weighted combination of positive gradients at spatial locations. These channel weights are then used to combine feature maps into a class-specific heatmap. Compared with LayerCAM’s location-dependent activation weighting, Grad-CAM++ provides a different aggregation of gradient information.

The method is included to examine whether highlighted regions remain similar when the weighting scheme changes. Both CAM methods are implemented using `pytorch-grad-cam` and target the same residual block, allowing their outputs to be compared without changing the model, input image, target class, or target layer. The maps are resized to the input resolution for display; this does not recover spatial detail absent from the underlying feature maps.

Occlusion [9] estimates regional influence by replacing a local input region with a reference value and measuring the change in the target-class output. Unlike the CAM methods, it does not require gradients or a selected internal layer. It is included to exam-

ine whether perturbing highlighted image regions changes the model output, providing a complementary input-based perspective.

The Captum implementation uses a 15×15 sliding window covering all three RGB channels, with a spatial stride of 8 pixels. The reference value is zero in normalized input space, corresponding to the ImageNet mean color rather than a black patch. Output differences from overlapping windows are averaged to obtain feature attributions.

Visualization and interpretation: for Occlusion, absolute attributions are summed across channels and min-max normalized to $[0, 1]$. Its displayed heatmap therefore represents influence magnitude without distinguishing regions that increase the target-class output from those that decrease it. The CAM maps and normalized Occlusion maps are overlaid on denormalized RGB images.

Comparisons focus on the location, concentration, and agreement of highlighted regions rather than absolute color intensity across methods. Apparent overlap with the visible optic-disc area is assessed qualitatively; it does not establish clinical validity or prove that the model has identified a disease-specific feature. Sample selection and the comparison protocol are described in Section 6.2.

5.3 Comparison Strategy

Classification results provide context through accuracy, macro F1-score, and confusion matrices. Since training configurations differ, performance differences are not attributed to architecture alone. The main qualitative comparison examines the three XAI methods on identical images and predicted classes using the criteria below.

Table 2: Criteria for qualitative XAI comparison.

Criterion	Analysis question
Location	Which image regions are highlighted, and do they visually overlap the optic-disc area or background?
Concentration	Is the explanation localized, diffuse, or fragmented?
Agreement	Which highlighted regions are shared across methods, and where do the methods disagree?
Prediction outcome	How do explanations differ between the selected correct and incorrect predictions?

Heatmap intensities are interpreted within each visualization, not as directly comparable numerical importance scores across methods. In particular, the displayed Occlusion map shows absolute attribution magnitude. Visual agreement alone does not establish explanation faithfulness or clinical validity.

6. Experimental Setup

The experiments compare classification performance across the three models, evaluate three XAI methods on a fixed ResNet50 checkpoint, and analyze misclassified test images.

Classification provides context for the explainability and error analyses.

6.1 Setup 1 - Baseline vs. Main Model

The scratch CNN, DenseNet121, and ResNet50 use the dataset partitions described in Section 4.2. Checkpoints are selected using validation loss and evaluated on the test set using accuracy, macro F1-score, and confusion matrices. Table 3 summarizes the model-specific settings.

Table 3: Classification training settings. LR denotes learning rate; CE denotes cross-entropy.

Setting	Scratch CNN	DenseNet121	ResNet50
Input size	224×224	224×224	320×320
Batch size	32	32	16
Optimizer	Adam	Adam	AdamW
Warm-up epochs / LR	—	$5 / 10^{-3}$	$5 / 10^{-3}$
Training / fine-tuning LR	10^{-4}	10^{-4}	Backbone: 10^{-5} ; head: 10^{-4}
Weight decay	10^{-4}	10^{-4}	10^{-3}
Training budget	Up to 30 epochs	After warm-up: 3 consecutive runs of up to 20 epochs each	After warm-up: up to 40 epochs per seed
Loss	Class-weighted CE	Unweighted CE	CE with label smoothing (0.1)
Stopping patience	8 epochs	5 epochs per run	7 epochs
Seeds	42	42	42, 123, 2026

ResNet50 results include the mean and standard deviation across three seeds, whereas the other models use one seed. DenseNet121’s consecutive fine-tuning runs are not independent repetitions. The models are trained on a Colab T4, RTX 5050 Laptop GPU, and RTX 5060 Laptop GPU, respectively.

6.2 Setup 2 - Main Research Experiment

The main experiment compares LayerCAM, Grad-CAM++, and Occlusion on the selected ResNet50 checkpoint from seed 2026. The model, input images, and target classes are held fixed across explanation methods. Implementation settings are provided in Section 5.2.3.

Sample selection. The test set is scanned without shuffling to collect the first image from each ground-truth class and the first three misclassified images. The class-based examples are not necessarily correctly classified, and the two groups may overlap. This deterministic selection is used for case-based inspection rather than a representative evaluation of explanations across the entire test set.

Comparison procedure. For each image, the original is displayed beside the LayerCAM, Grad-CAM++, and Occlusion overlays, together with the true and predicted labels. All three explanations target the predicted class, allowing incorrect decisions to be examined directly. Visual comparison follows four questions:

1. Which regions are highlighted, and do they visually overlap the optic-disc area or background?
2. Are the highlighted regions concentrated, diffuse, or fragmented?
3. Where do the three methods agree or disagree?
4. What differences appear between the selected correct and incorrect predictions?

The analysis describes observable patterns and considers possible explanations for model errors. These interpretations are treated as hypotheses rather than verified causes. No expert localization annotations or quantitative faithfulness measurements are used, so the experiment does not establish clinical validity or rank the methods by explanation accuracy.

6.3 Setup 3 - Error Analysis Experiment

This experiment examines the first three misclassified test images encountered during the dataset scan. The same ResNet50 checkpoint and XAI settings as in Setup 2 are used, with explanations targeting the incorrectly predicted class.

For each case, the true and predicted labels are displayed alongside the original image and the three explanation maps. The analysis examines whether highlighted regions overlap the visible optic-disc area or unrelated regions, and whether the methods agree. Potential explanations for the errors are discussed as hypotheses rather than verified causes. Detailed cases are presented in Section 8.

7. Results and Discussion

7.1 Classification Results

Table 4 reports the selected stored test-result records. The custom CNN has substantially lower accuracy and macro-F1 than the two pretrained models. In the selected records, DenseNet121 has the largest single-run accuracy and macro-F1, while the ResNet50 multi-seed study reports close results across three seeds. These comparisons must be interpreted cautiously because CNN/DenseNet and ResNet50 use documented input sizes of 224 and 320, respectively, and the cross-split duplicates can inflate all test estimates.

For the XAI checkpoint (ResNet50 seed 2026), per-class F1 is 0.9242 for normal, 0.8436 for early, and 0.9245 for advanced. The three-seed summary similarly reports early as the lowest mean-F1 class (0.8344 ± 0.0091), compared with normal (0.9192 ± 0.0052) and advanced (0.9205 ± 0.0051). Thus, in the available results, the early class is the more difficult class. This is consistent with a more challenging decision boundary, but the experiments do not establish the cause of those errors.

Table 4: Selected stored classification results on the within-split-deduplicated test set.

Model/result record	Accuracy	Macro-F1	Interpretation
Custom CNN (<code>cnn_scratch_v2</code>)	0.6699	0.6481	From-scratch baseline
DenseNet121	0.9099	0.8981	Selected stored DenseNet result
ResNet50, seed 2026	0.9083	0.8974	Checkpoint used for XAI
ResNet50, 3-seed mean $\pm$ SD	0.9031 ± 0.0047	0.8914 ± 0.0053	Seeds 42, 123, 2026

The seed-2026 confusion matrix in Table 5 provides the counts underlying this result. Its most frequent off-diagonal entry is normal predicted as early (32 cases), followed by normal predicted as advanced (29 cases). Early-class errors are split between normal (16) and advanced (19), while advanced is relatively often identified correctly (392 of 408 in the saved matrix). These counts describe the stored test evaluation only and are not a clinical error analysis.

Table 5: ResNet50 seed-2026 test confusion matrix. Rows are true classes; columns are predicted classes.

True \ predicted	Normal	Early	Advanced
Normal	512	32	29
Early	16	205	19
Advanced	7	9	392

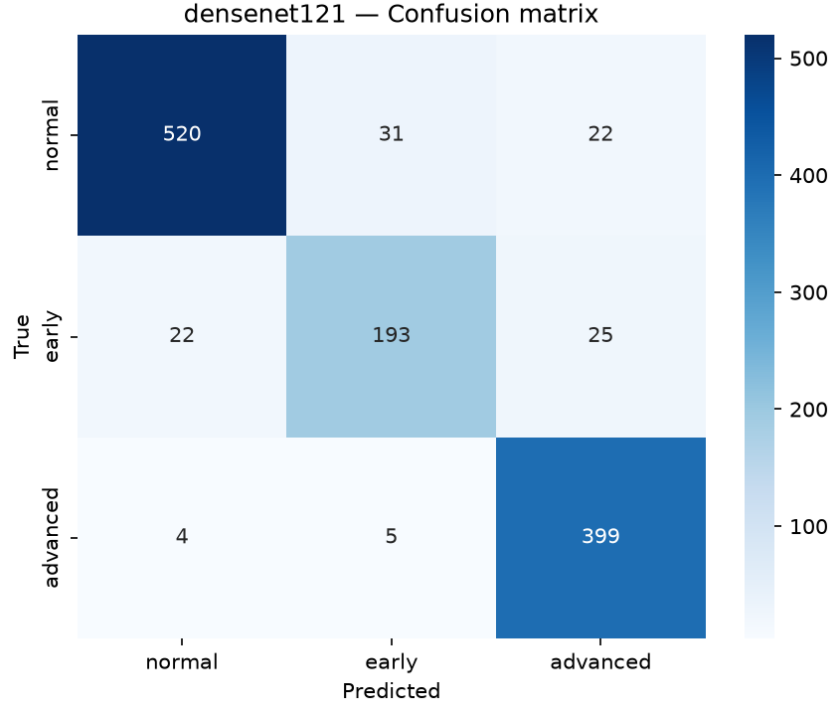


Figure 1: DenseNet121 confusion matrix for the selected stored DenseNet result. This supports the classification comparison in Table 4.

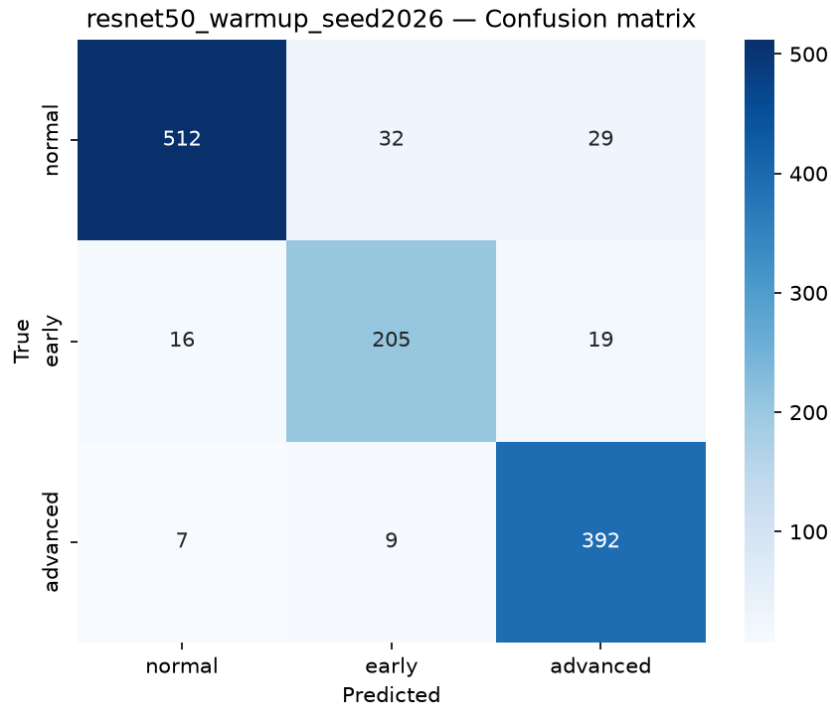


Figure 2: Confusion matrix for the selected ResNet50 seed-2026 checkpoint.

7.2 Model Selection

ResNet50 seed 2026 was selected for the XAI pipeline because the checkpoint is explicitly saved, downloaded, and used by the XAI notebook, and because its result is part of a

three-seed stability study. Its selected checkpoint has test accuracy 0.9083 and macro-F1 0.8974. It should not be described as unequivocally the numerically best architecture in every stored file: the selected DenseNet121 record is slightly higher, and the comparison conditions are not fully matched. Rather, ResNet50 is the model actually selected for the explanation experiment, with a documented 320×320 preprocessing path and reproducible checkpoint provenance.

`results/metrics/resnet50.json` reports an unannotated ResNet50 result (accuracy 0.9124, macro-F1 0.9035) that conflicts with the documented warm-up multi-seed study. It is not used for model selection in this report; its experiment configuration should be identified before submission.

7.3 XAI Results

Figure 3 presents the actual class-representative grid. In the available figure, LayerCAM and Grad-CAM++ show closely aligned high-response patterns across the displayed examples, generally with the strongest response around the visible optic-disc area. Occlusion displays a visibly coarser, patch-like attribution pattern and, in some rows, emphasizes local areas that differ from the gradient-based maxima. This is an observation about the rendered maps, not evidence that a particular anatomical feature caused a clinically valid diagnosis.

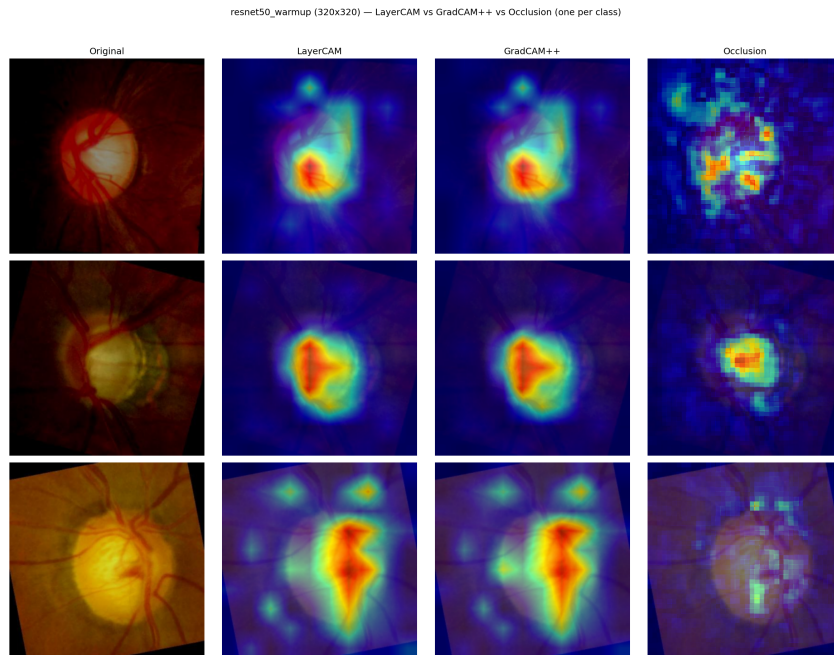


Figure 3: Class-representative XAI comparison generated by `xai_analysis.ipynb`. The notebook selects the first encountered test image for each ground-truth class.

The available grid enables a method-level comparison but no numerical claim about localization accuracy, agreement, or faithfulness. LayerCAM and Grad-CAM++ appear smoother and more spatially concentrated than the patchwise occlusion maps. Their agreement is strongest in the central disc-adjacent patterns visible in the displayed class-

representative examples; Occlusion provides a complementary perturbation perspective, but its broad and blocky regions make pixel-level correspondence inappropriate. Additional randomly sampled and clinically annotated examples would be needed to estimate how often these patterns occur.

7.4 Comparison of XAI Methods

Table 6 summarizes the implemented methods and the cautious qualitative findings supported by the two generated grids. The same predicted class is used for all maps; hence differences reflect the explanation procedure, not a different decision target. Agreement between methods is evidence of a shared visual pattern under this implementation, not proof that the model uses an anatomically correct feature.

Table 6: XAI comparison under the fixed ResNet50 seed-2026 protocol.

Method	Information used	Observed visual behavior in saved grids	Limitation
LayerCAM	Activations and class gradients at ResNet50 <code>layer4[-1]</code>	Smooth disc-adjacent high-response regions; visually similar to Grad-CAM++	Qualitative only; no localization ground truth
Grad-CAM++	Class-specific derivatives and activation maps at the same layer	Similar broad maxima to LayerCAM in displayed rows	Resolution limited by target-layer features
Occlusion	Change in target output after patch replacement	Coarser, patch-like attribution; sometimes emphasizes different local patches	Depends on zero baseline, 15×15 window, and stride 8

8. Error and Qualitative Analysis

The separate error grid provides the most direct evidence for analysing model failures. The protocol selects the first three misclassified test images. All three have ground-truth label advanced; their predicted labels are, in order, early, normal, and early. Thus, the available sample is not a balanced catalogue of all error types. It demonstrates that the selected model can under-predict advanced cases, but it does not establish a population-wide error rate beyond the complete confusion matrix.

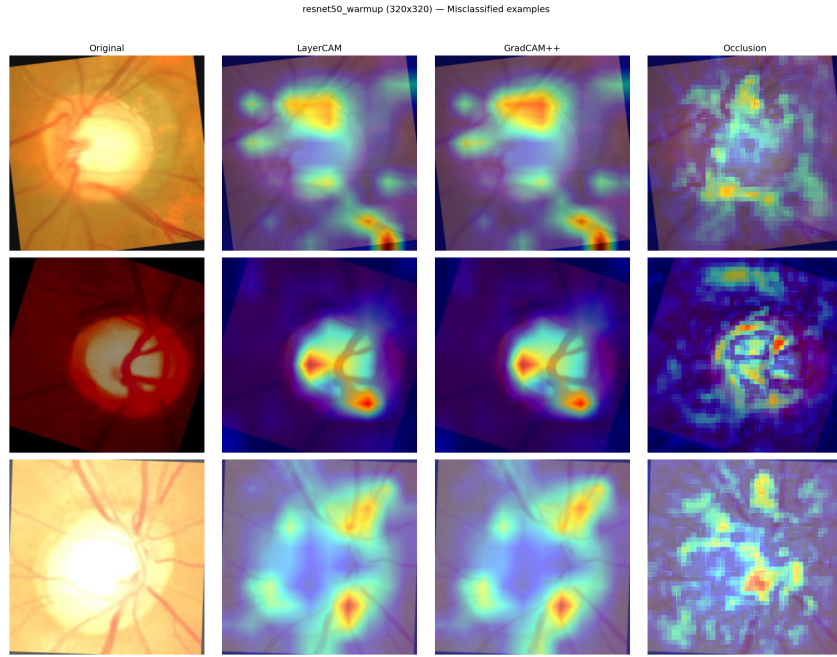


Figure 4: Misclassified examples generated by `xai_analysis.ipynb`: advanced $\rightarrow$ early, advanced $\rightarrow$ normal, and advanced $\rightarrow$ early.

9. Conclusion and Limitations

9.1 Limitations

- **Data leakage:** Cross-split duplicates (459 training-validation and 358 training-test shared hashes) may inflate evaluation scores.
- **Comparison conditions:** Different input resolutions and unresolved configuration discrepancies limit direct comparisons between models.
- **Evaluation scope:** XAI analysis used a small, non-random sample without quantitative localization or faithfulness evaluation. No external validation dataset was used.

9.2 Conclusion

The selected results show that pretrained DenseNet121 and ResNet50 outperformed the custom CNN under the evaluated configurations. ResNet50 showed limited variation across three seeds, while the early class consistently had the lowest F1-score. These findings remain subject to the limitations above.

Using ResNet50 seed 2026, LayerCAM and Grad-CAM++ produced visually similar attribution patterns, whereas Occlusion provided coarser, patch-based maps. The analyzed misclassifications show that similar explanations can accompany incorrect predictions. The heatmaps therefore support exploratory model analysis but do not establish clinical validity.

References

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Appendix

Table 7: Member contributions

Member	Contribution
Nguyen Minh Nhat	Presentation; collected demo images; ran and tested the demo notebook across all sourcing modes; wrote the Setup 3, Comparison of XAI Methods, and Error and Qualitative Analysis sections
Hoang Thu An	Verified the README install and run instructions on a fresh clone; wrote the Abstract, Introduction, and Conclusion sections; compiled, formatted, and proofread the final report
Hoang Ngan Anh	Implemented shared codebase (dataset pipeline, common utilities); built and trained the from-scratch CNN; wrote the Dataset and Data Preparation, Baseline CNN Method, and Setup 1 sections
Tran Mai Anh	Built the ResNet50 model; ran the XAI analysis (LayerCAM, Grad-CAM++, Occlusion); wrote the Selected Transfer-Learning Model, Explainability Methods, Model Selection, and XAI Results sections
Vu Duc Quang	Trained the ResNet50 model (3-seed experiment); verified and saved the XAI output figures; wrote the Fine-Tuning Strategy, Setup 2, and Classification Results sections
Doan Quoc Viet	Built and trained the DenseNet121 model; wrote the Related Work, Comparison Strategy, and Limitations sections; completed the Appendix and verified the submission checklist